

University of South Carolina
College of Engineering & Computing
Department of Mechanical Engineering

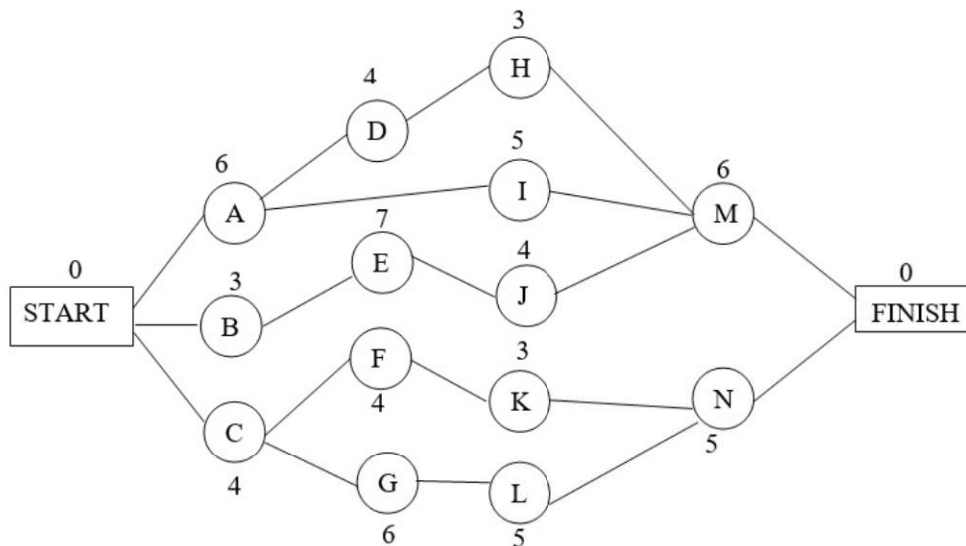
EMCH380 – Project Management

Homework 2

Due date: Thursday, September 30 (by 2:49pm)

- Notes:
- Homework should be submitted on Blackboard, hard copies will not be accepted.
 - SafeAssign will be used to identify areas of overlap between the submitted assignment and existing works.

1. Ken Johnston, the data processing manager for Stanley Morgan Bank, is planning a project to install a new management information system. He now is ready to start the project and wishes to finish in 20 weeks. After identifying the 14 separate activities needed to carry out this project, as well as their precedence relationships and estimated durations (in weeks), Ken has constructed the following project network:



(25
points)

- Find all the paths and path lengths through this project network. Which of these paths is a critical path?
- Find the earliest times, latest times, and slack for each activity. Will Ken be able to meet his deadline if no delays occur?
- Use the information from part (b) to determine which of the paths is a critical path. What does this tell Ken about which activities he should focus most of his attention on for staying on schedule?

2. For the precedence network information given below:

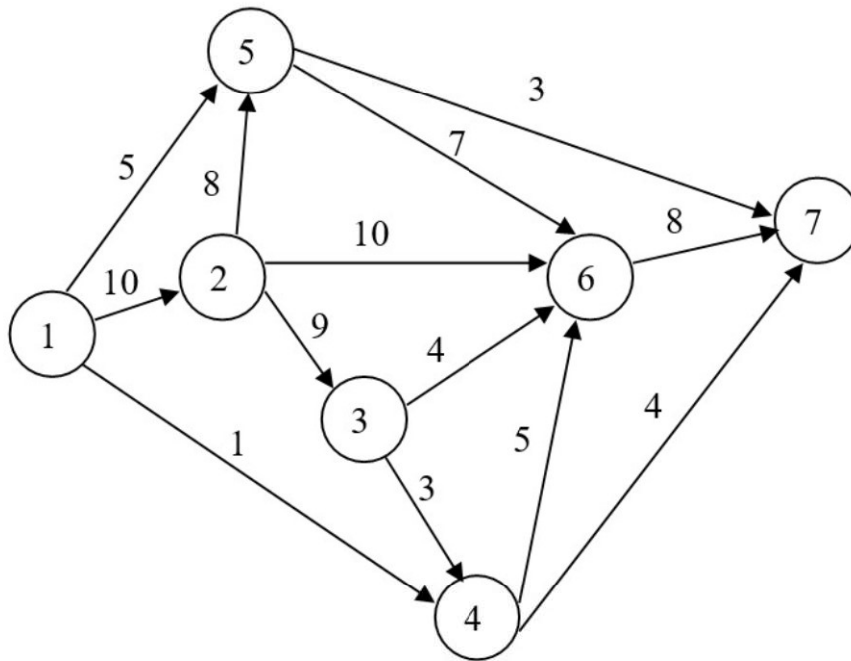
- Set up a table, and without drawing the network diagram, compute the schedule dates (ESD, EFD).
- Compute the lag for each link and identify activities on the critical path.
- Compute the four floats (TF, FF, INTF, INDF) for each activity.

i	ACT	DUR	PREDECESSORS
5	A	2	-
10	B	3	A
15	C	6	A
20	D	5	A
25	E	9	B
30	F	7	B,C
35	G	2	C
40	H	4	D
45	J	3	E,F,G
50	K	5	F,G,H
55	L	2	G,H
60	M	7	J,K,L
65	N	4	K,L
70	P	2	M,N

(25
points)

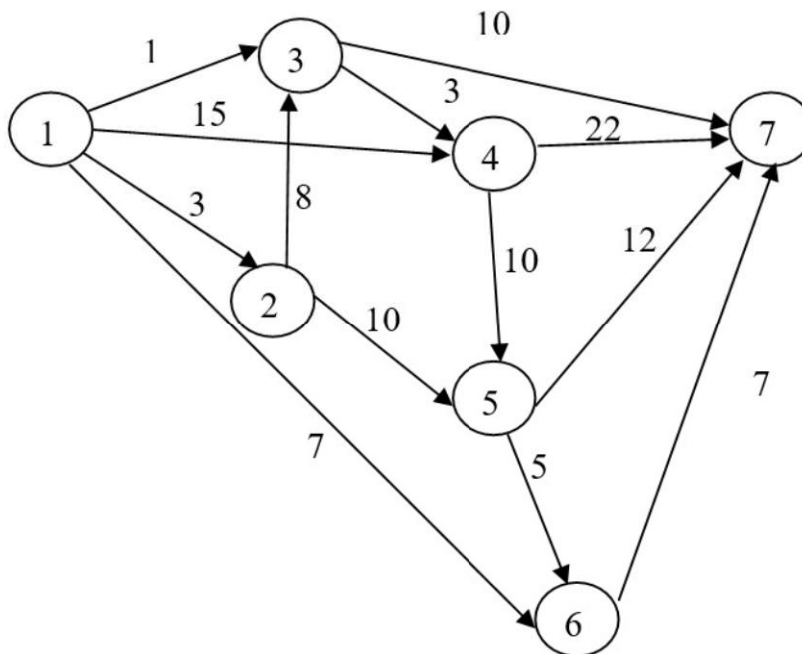
3. Determine the critical path for each of the project networks shown below. Compute the four floats for each of the activities in the network.

a. **Network #1:**



(25
points)

b. **Network #2:**



4. For the activity on arrow network information given below:

- a. Set up a table and without drawing the network diagram, compute the four schedule dates (ESD, EFD, LSD, LFD) and the four floats (TF, FF, INTF, INDF).
- b. Identify the activities on the critical path and the project duration.

i	j	ACT	DUR
5	10	S	4
5	15	R	3
10	20	A	5
10	25	E	2
10	30	d1	0
15	30	G	2
15	35	B	1
20	40	d2	0
20	60	L	2
25	40	K	2
25	45	d3	0
30	45	C	3

i	j	ACT	DUR
30	50	H	6
35	45	d4	0
35	55	J	4
40	65	M	6
45	65	d5	0
45	70	T	3
50	65	W	3
50	70	d6	0
55	70	X	5
60	75	N	6
65	75	P	4
70	75	V	4

(25
points)